ALPHALIZE TOOLS MANAGEMENT

Device Login Configuration

Odoo Module — Technical & User Manual

Module Version: 19.0.3.0.0
Odoo Compatibility: Odoo 19
Category: Technical / Authentication
Date: April 2026

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1. Module Overview

The Device Login Configuration module enforces MAC address-based device approval for the Tool Management mobile app. Only pre-registered and approved devices are permitted to connect to Odoo and authenticate users.

Module Name	Device Login Configuration
Technical Name	`device_login_config`
Version	19.0.3.0.0
Category	Technical / Authentication
Dependencies	web, base
License	LGPL-3

1.1 Purpose

This module controls which physical devices are allowed to access the Odoo-based Tool Management mobile application. By enforcing device-level registration (using a unique Device ID or MAC address), administrators can ensure that only authorized tablets or mobile devices can connect to the system — preventing unauthorized access even if a user has valid login credentials.

1.2 Key Capabilities

- Pre-register devices by their Device ID (UUID, MAC address, or alphanumeric ID) before they ever connect
- QR code-based registration: display a QR on the Odoo form, scan it from the app to auto-fill Device ID and Name
- Three-status lifecycle management: Pre-Registered → Active → Blocked
- Block specific devices at any time to immediately revoke access
- Redirect unregistered devices from the login page to the device setup screen
- Full REST/JSON-RPC API for the mobile app integration
- Access control: only System Administrators can manage device records

2. Installation

2.1 Prerequisites

- Odoo 19 (or compatible version) with admin access
- Python package: qrcode (for QR code generation)

- The module zip or folder placed in your Odoo addons directory

2.2 Installation Steps

1. Copy the device_login_config folder into your Odoo custom addons path
2. Restart the Odoo service to detect the new module
3. Log in to Odoo as an Administrator
4. Navigate to Settings → Activate Developer Mode
5. Go to Apps → Update Apps List
6. Search for Device Login Configuration and click Install
7. After installation, the Device Registry menu appears in the top navigation bar (visible to System Administrators only)

2.3 Python Dependency

The QR code generation feature requires the qrcode Python package. Install it on your Odoo server:

```
pip install qrcode[pil]
```

If the package is not installed, the QR code field on device forms will silently return empty — all other module features continue to work normally.

3. Device Registry — Data Model

The core data model is device.registry. Each record represents one physical device.

3.1 Fields

Field Name	Label	Type	Description
`mac_address`	Device ID	Char	Unique device identifier (UUID, MAC, or alphanumeric 4–64 chars). Auto-filled on QR scan.
`device_name`	Device Name	Char	Human-readable device label. Auto-filled from device model on QR scan. Defaults to Pending.
`status`	Status	Selection	Lifecycle state: pre_registered, active, or blocked. Default: pre_registered.
`device_id`	Device UUID	Char	UUID generated by the app and linked on first activation. Read-only.
`base_url`	Base URL	Char	Server URL this device connected to. Filled on activation. Read-only.

`database_name`	Database Name	Char	Odoo database this device is connected to. Filled on activation. Read-only.
`last_login`	Last Login	Datetime	Timestamp of last successful device login or activation. Read-only.
`registration_qr_image`	Registration QR Code	Binary	Computed QR code image. Encodes record ID + database for scan-to-register flow.

3.2 Device Status Lifecycle

Status	Color Indicator	Meaning
Pre-Registered	Orange/Warning	Device record created by admin but app has not yet scanned/activated
Active	Green/Success	Device has been activated by the app and is permitted to log in
Blocked	Red/Danger	Device is explicitly blocked; all access attempts will be denied

4. Administrator Workflows

4.1 Registering a New Device (QR Method — Recommended)

This is the primary registration method. The admin creates a record in Odoo, which generates a QR code that the device app scans to self-register.

8. Navigate to Device Registry in the top menu bar
9. Click New to create a new device record
10. A QR code is generated automatically and displayed on the form
11. On the tablet or mobile device, open the Tool Management app
12. Tap Configure Device, then tap Scan QR
13. Point the device camera at the QR code displayed in Odoo
14. The app automatically fills in the Device ID and Device Name fields
15. Reload the Odoo form page to see the registered device details populated
16. The device status is set to Active automatically after scanning

4.2 Registering a Device Manually

If QR scanning is not available, the admin can enter the Device ID manually.

17. Navigate to Device Registry and click New
18. Enter the Device Name (a descriptive label for the device)
19. Enter the Device ID in one of the accepted formats:

- UUID: e.g., 550e8400-e29b-41d4-a716-446655440000
- MAC Address: e.g., AA:BB:CC:DD:EE:FF or AA-BB-CC-DD-EE-FF
- Plain alphanumeric ID: 4–64 characters, letters, digits, hyphens, underscores

20. Save the record — status is set to Pre-Registered automatically
21. The device will be permitted to connect once it activates via the app

4.3 Blocking a Device

22. Go to Device Registry and open the target device record
23. Change the Status field to Blocked
24. Save the record

The device is immediately denied access. Any login attempt from that device will be rejected, regardless of user credentials.

4.4 Viewing All Devices

The Device Registry list view shows all registered devices with the following columns: Device Name, Device ID, Database Name, Base URL, Status (color-coded), and Last Login.

Use the search bar to filter by:

- Device ID or Device Name
- Database Name
- Status (Pre-Registered / Active / Blocked)
- Group by Status or Group by Database using the filter options

5. Mobile App Workflow

5.1 First-Time Device Setup

When a device opens the Tool Management app for the first time (or after clearing app data), it is redirected to the Device Setup screen.

25. Open the app — the Device Setup screen appears automatically
26. Enter the Server URL (base URL of the Odoo server)
27. Select the Database from the dropdown list (populated via /device/databases)
28. The app reads the device's unique UUID automatically
29. Tap Scan QR if the admin has prepared a QR code on the Odoo form
30. Or enter the MAC Address / Device UUID manually if QR is unavailable
31. Tap Connect — the app calls POST /device/lookup to check registration status
32. If the device is pre-registered, the app calls POST /device/activate to link the UUID
33. Status changes to Active on the Odoo server

34. The app stores a session cookie (odoo_device_id) and redirects to the login screen

5.2 Subsequent Logins

Once a device is activated, it stores its Device UUID in a cookie (odoo_device_id, valid for 30 days). On each visit to the Odoo login page:

35. The controller reads the cookie and validates the Device UUID
36. If the UUID is found in the device.registry with status Active, the login page is shown
37. If the UUID is not found or the device is Blocked, the user is redirected to /device/config for re-setup

5.3 Access Denial Scenarios

- Device UUID not in the pre-registered list — redirected to setup screen
- Device status is Blocked — activation is denied with a blocked message
- Different app installation tries to claim an already-Active record — returns already_active status

6. API Endpoint Reference

All endpoints are accessible without authentication (auth='none') to allow the mobile app to connect before any user is logged in. All JSON-RPC endpoints use POST method.

Endpoint	Method	Auth	Description
`/device/config`	GET	None	Device setup page — shown to unregistered devices
`/device/lookup`	POST	None	Check if a Device ID is pre-registered (JSON-RPC)
`/device/activate`	POST	None	Activate a pre-registered device, link UUID (JSON-RPC)
`/device/register-from-scan`	POST	None	QR scan registration — updates specific record (JSON-RPC)
`/device/databases`	POST	None	Returns list of available Odoo databases (JSON-RPC)
`/device/init`	POST	None	Unified startup endpoint — check or auto-register (JSON-RPC)
`/device/register`	POST	None	HTTP form-based device registration
`/device/check`	POST	None	Check if a device UUID is registered (JSON-RPC)

`/device/registration-qr.png`	GET	User	Returns QR code PNG for the current session database

6.1 POST /device/lookup

Checks whether a Device ID (mac_address) is pre-registered in the system. This is Step 1 of the app setup flow.

Request Parameters:

- mac_address (string, required) — The device's unique identifier
- database_name (string, required) — The Odoo database to check against

Response — Device Found:

```
{ "status": "found", "device_name": "Reception Tablet", "device_status": "pre_registered" }
```

Response — Device Not Found:

```
{ "status": "not_found", "message": "This device is not registered. Please contact your administrator." }
```

6.2 POST /device/activate

Links the app UUID to a pre-registered record and marks the device as Active. This is Step 2 of the app setup flow.

Request Parameters:

- mac_address (string, required) — Device identifier matching the pre-registered record
- database_name (string, required) — The target Odoo database
- device_id (string, required) — App-generated UUID to link to this record
- base_url (string, optional) — Server URL used by this device

Response Values:

- activated — Device successfully activated; status set to Active
- already_active — Device was already active; last_login timestamp updated
- blocked — Device is blocked by the administrator; access denied
- not_found — No matching pre-registered record found
- error — Server-side error occurred

6.3 POST /device/register-from-scan

Used when the app scans the QR code displayed on the Odoo Device Registry form. Updates the specific record whose ID is encoded in the QR code.

Request Parameters:

- device_id (string, required) — The app UUID read by the device
- device_name (string, optional) — Device model name from the app
- database_name (string, required) — The target database
- base_url (string, optional) — Server URL
- record_id (integer, optional) — The Odoo record ID from the QR code payload

The endpoint follows this resolution order:

38. If record_id is provided, update that specific record
39. If no record_id, check if a record with this device_id already exists and update it
40. If no existing record, claim the oldest unclaimed Pre-Registered (Pending) record
41. As a last resort, create a new device record

6.4 POST /device/databases

Returns all available databases on the Odoo server. Used by the app to populate the database picker during setup.

```
Response: { "databases": ["mydb", "testdb", "production"] }
```

6.5 POST /device/init

A unified startup endpoint. Checks if the device is already registered; if not, auto-registers it and returns the appropriate status.

- registered: true — Device exists; last_login updated
- registered: false, just_registered: true — New device; auto-registered successfully
- registered: false, error: "..." — Validation or server error

7. Access Control & Security

7.1 Permission Matrix

User Group	Read	Write	Create	Delete
System Administrator	Yes	Yes	Yes	Yes
Regular User	Yes	No	No	No

7.2 Menu Visibility

The Device Registry menu and all its sub-items are restricted to the System Administrator group (base.group_system). Regular users do not see this menu in the navigation bar.

7.3 Device ID Validation

When a Device ID is entered manually, it is validated against three accepted formats:

- UUID format: 8-4-4-4-12 hexadecimal characters (e.g., 550e8400-e29b-41d4-a716-446655440000)
- MAC address: XX:XX:XX:XX:XX:XX or XX-XX-XX-XX-XX-XX format
- Plain alphanumeric: 4 to 64 characters consisting of letters, digits, hyphens, and underscores

The mac_address field also has a unique constraint — the same Device ID cannot be registered more than once on the same database.

8. Module File Structure

File / Folder	Purpose
`__manifest__.py`	Module metadata: name, version, dependencies, assets, description
`models/device_registry.py`	Core data model (device.registry): fields, QR generation, validation
`controllers/device_controller.py`	HTTP/JSON-RPC controllers: login override, /device/* endpoints
`controllers/main.py`	Additional endpoints: /device/lookup, /device/activate, /device/register-from-scan, QR PNG
`views/device_registry_views.xml`	List, form, search views; menu items and action window
`security/device_security.xml`	Security category and groups definition
`security/ir.model.access.csv`	Model-level access rights (admin: full, user: read-only)
`static/src/js/device_config.js`	Frontend JavaScript for the device setup page
`static/src/js/device_qr_scanner.js`	QR code scanner integration for the backend UI
`static/src/js/device_autosave.js`	Auto-save behavior for device configuration forms
`static/src/css/device_qr_scanner.css`	Styling for the QR scanner overlay
`templates/device_config.xml`	QWeb template for the /device/config setup page

9. Troubleshooting

9.1 QR Code Not Displaying

- Ensure the qrcode Python package is installed on the Odoo server
- Run: pip install qrcode[pil] and restart the Odoo service
- The QR code is only generated after the record is saved (requires a record ID)
- Reload the form after saving if the QR does not appear immediately

9.2 Device Redirected to Setup Page on Every Login

- The device cookie (odoo_device_id) may have been cleared — re-register the device
- Confirm the device record has status Active in Device Registry
- Verify the correct database is selected during device setup

9.3 Device Cannot Activate — Returns not_found

- The Device ID entered on the app must exactly match the mac_address field in the Odoo record
- Check for leading/trailing spaces in both the app input and the Odoo record
- Use the QR scan method to avoid manual entry errors

9.4 Device Returns blocked Status

- An administrator has set the device status to Blocked in Device Registry
- Contact the system administrator to change the status back to Active

9.5 Database List Empty on App Setup Screen

- The /device/databases endpoint relies on odoo.service.db.list_dbs — ensure the Odoo DB manager is accessible
- Check Odoo server logs for errors related to this endpoint

10. Version History

Version	Changes
19.0.3.0.0	QR-based registration with record ID; /device/init unified endpoint; /device/register-from-scan with 4-step resolution logic; flexible Device ID validation (UUID/MAC/alphanumeric)

This document was auto-generated from the device_login_config module source code.

Alphalize Tools Management — April 2026